
Creating spatial analysis tools for archaeology

WITH PYTHON AND QGIS

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Chapter 1

Project datasheet

Project Title	Creating spatial analysis tools for archaeology
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Chapter 2

Introduction and background

2.1 Introduction

The Netherlands has gained a substantial amount of data on the distribution of Roman settlement in the past 12 years. However, all this data is recorded in different databases with differences in quality. Regional en supra-regional analysis of this data is extremely valuable for the understanding of the effects and settlement development dynamics by force of Roman imperialism in the Lower German Limes. While this project uses data from ARCHIS (Rijksdienst voor het Cultureel Erfgoed, 2026), PAN (Portable Antiquities Netherlands (PAN), 2026), specific inventories, and archaeological project reports concerning the Roman time period in the Netherlands, it will not be explaining the archaeology itself as this project is focused on created a digital product that can be used in further research.

2.2 Background: Roman Netherlands

The Romans had a significant presence in the Netherlands as they were here for almost five centuries. Julius Caesar first reached the Lower Rhine around the middle of the first century BC, but the actual incorporation of Gaul into the Empire happened after 31 BC, when his adopted son became the sole ruler. Rome almost immediately started building up a military presence along the fringes of Gaul, including along the Rhine. By AD 80s, the provinces of Upper and Lower Germany were created. The southern part of the Netherlands belonged to the Lower Germany (Germania Inferior) province. The Lower Rhine became the northern Limes (border) of the Empire.

Since the border was running along the river, there are many sites and finds

related to the military that kept watch over it. Examples of these are castella, watchtowers and the Limes road. However, not only the military was living in these places at the time. Also the peoples of the tribes living here before the Romanization continued to live here, and interacted with peoples of tribes beyond the Limes. All this left its traces in the ground, and the Dutch waterlogged soil of the river delta kept finds in good condition. An overview of all the evidence and progress since 2000 to roughly 2020 had been compiled and made available by Hessing et al. (2021). New research is still being completed, with one of the more recent publications being De Soto et al. (2025), where an overview of the roads is given.

The Romans stayed in the Netherlands until the fall of the Empire, which meant they were here for almost five centuries. Graafstal et al. (2021) wrote a report to try to get the Lower German Limes accredited as UNESCO world heritage. This was successful, and in 2026 heritage sites celebrated 5 years of Limes as world heritage.

2.3 Technical background

This section gives some background on the technical side of this project, mostly related to the files used inside the QGIS environment.

2.3.1 Shapefile

A shapefile is a format for storing geospatial vector data. A vector feature can consist of points, lines, and/or polygons. Each feature has attributes attached to it that describe it. Examples of attributes are the name or what it represents. While referred to as 'shapefile' in conversations, it is actually a collection of files. Three of these are mandatory, and they have the file extensions of '.shp', '.shx', and '.dbf'. The file names will be the same, with only the extension being different. The file with '.shp' is the actual shapefile itself and stores the feature(s) geometry. The file with '.shx' stores the positional index of the feature geometry, while the '.dbf' file stores the attributes.

2.3.2 GeoPackage

A GeoPackage is a single file that has the filename extension '.gpkg'. It is a SQLite 3 database with a specific schema that defines data and metadata tables. It describes a set of requirements for storing vector features, tile matrix sets and raster maps at various scales. Its initial release was in February 2014 and it is

open format. GeoPackage was designed to be as lightweight as possible and to have a single file. This is its main advantage over shapefiles, which need at least three files.

Chapter 3

Aims and justification

The specific aim for this project was to adapt and/or create new tools to characterize spatial units based on specific requirements. This was done as there is a lack of established methods to aggregate at the appropriate spatial resolution. So far percolation is used as a method of spatial clustering at multiple scale levels (Maddison & Schmidt, 2020), however this is purely based on geographical distance of the sites and not with find information categories that play a big part in archaeology.

The main aim for this project is thus combining find information with the spatial data of the site it belongs to, and start spatial analysis using visualizations of not only sites, but also finds. The sub-aim is to create a tool that could be adapted and expanded on if the results of the main aim are sufficient. This first iteration will be specific to Dutch archaeology, as the structure of data is standardized if exported from the national archive ARCHIS.

The results of this would ideally be used in further research by academics on the Roman frontier, but possibly on other archaeological periods as well.

Chapter 4

Research design

4.1 Approach

Archaeological data is complex. It deals with spatial information on the larger scale of site placement and on the small scale of finds within a site (Renfrew & Bahn, 2020, fig 2.1, pp 49). These finds also have very detailed information attributed to them, including but not limited to material type, sub-type, classification and dating. When things get difficult, we as humans tend to make tools to help us overcome the difficulty. Just as with physical artifacts, that archaeologists then can find during field work, we also create cognitive artifacts. These cognitive artifacts manage to overcome the limits of our brainpower (Norman, 2005, pp. 127). Humankind has invented many a tool to accomplish this, both physical tools, as mental tools. Examples of physical tools are paper and pencil, to write things down and free up working memory by not having to keep reminding ourselves of things; while an example of a mental tool is language, as its structure and rules help to communicate better with other humans. (Norman, 2005, pp. 4).

One way to make sense of spatial data is by visualizing it on a created representation of the area studied. Norman (2005, pp. 127) also describes humans as "excellent perceptual creatures", meaning that we can see a pattern and understand it almost immediately. Card et al. (2007) describe, according to Fekete et al. (2008), visualization as "the use of computer-supported, interactive visual representation of data to amplify cognition". This together then forms the basis for information visualization systems that Fekete et al. (2008) explain: using human perception as a fast filter, as a perceived pattern might hint at a pattern in the actual data. In that way an information visualization can help people to narrow data down from a large data set and focus on parts that seem more interesting.

The visualization of spatial data is usually done with maps and thus they have a long history of being used to recognize patterns into spatial data. An early and famous example is a map from central London in 1854 where Dr. John Snow plotted the cholera deaths and water pumps in the area and came to the conclusion that the deaths nearly all occurred from people who used the Broad Street pump (Tufte, 2001, pp.24, Fekete et al., 2008, pp.9). Using computers and photographs the density of information has increased tremendously. The map shown on page 27 of Tufte (2001) shows the distribution of 1.3 million galaxies in the northern galactic hemisphere. This shows that data maps can show millions of bits of information on a limited surface, which gives maps an incredible powerful way to display things that is not confined to a single discipline.

As mentioned before, archaeological data is complex because of the spatiality on different scales and added attributes. Since each site has its own complexities, many of the solutions to explain the data could be classified as ad hoc solutions. Some standardization might have happened using Geographical Interface Systems (GIS) to help with visualizing spatial data. Inside a GIS environment, the user is able to plot the data points on a coordinate reference system (CRS). This is also helpful when trying to combine data from traditional archaeology with the data found in specific databases like the one found on the website of Portable Antiquities Netherlands (Portable Antiquities Netherlands (PAN), 2026) or the Archaeological Information system (ARCHIS) from the Cultural Heritage Agency of the Netherlands (Rijksdienst voor het Cultureel Erfgoed, 2026). However, since GIS environments were not intended for analysis of bigger data sets, processes can become a bit slow. To circumvent this, the use of the Python programming language was decided upon. The output of this project will thus be multiple Python files with code, and this report. Depending on which functions within the Python code is run, the output of those consists of generated images, created 'comma separated value' files or both.

4.2 Methodology

This project was finished to completion by creating scripts in Python. The results of the searches were cross-referenced with the original data files and the locations on the plots were checked against the data in QGIS.

4.3 Sources

4.3.1 Literary sources

Non-data set resources were found online through the Vrije Universiteit (VU) and Universiteit van Amsterdam (UvA) libraries. Starting point was a blog post on Archaeoinformatics about percolation (Schmidt, 2021). Other literature regarding archaeology was found searching for “archaeology statistics” and “archaeology spatial analysis”. This resulted in the books by Drennan (2010) and Gillings et al. (2020), which then formed the basis for further search. This resulted in the articles by Bevan (2012), Bevan et al. (2013), and Cooper and Green (2017) were found.

Literature about archaeological practices (Renfrew & Bahn, 2020), the cognitive artifacts (Norman, 2005) and visualizations (Fekete et al., 2008; Tufte, 2001) as described in section 4.1 were part of the authors (previous) studies and were provided by her teachers.

4.3.2 Data sources

The data sources used in the project were mostly provided by M. R. Groenhuizen and P. Verhagen. These include the original ‘comma separated value’ (csv) files with the Roman data, and the palaeogeography shapefile. The csv files are part of the ongoing project “Simulating Globalisation in the Roman Empire” and contain data from ARCHIS (Rijksdienst voor het Cultureel Erfgoed, 2026) as the author does not have direct access. The palaeogeography file is part of the ongoing project “Constructing the Limes”, and the first part was constructed by Groenhuizen (2018) during his PHD project. The provincial boundaries of the Netherlands was found on PDOK (Centraal Bureau voor de Statistiek (CBS), n.d.).

4.3.3 Potential biases in the sources

Since the data has been collected and/or created by others, there is the chance for biases in there, as there is not an easy way to verify the data. That is on top of the bias that archaeological data inherently has, as not all has survived the passing of time, being found or being registered.

Chapter 5

Workflow and process

5.1 Data acquisition, processing and analysis

5.1.1 Software used

Python scripts were written in PyCharm Community Edition version 2025.2.41, with Python version 3.13. Table 1 shows all the packages that were installed. The main packages used were Pandas, GeoPandas, and matplotlib. GIS operations were performed in QGIS version 3.44.7.

5.1.2 Analytical techniques used

This project uses mostly visual techniques, but the results could be easily used in statistical analysis as well. In the *statistics.py* script, there is code provided for a Chi square test.

5.1.3 Justification for chosen techniques

While it seems logical to put more statistical analysis already within the python file, it was omitted this time as different research questions require different statistical analysis to be done and it is impossible to know which ones will be used in future. This is also complicated by the fact that some future researchers might first do some extra data handling outside of the python environment, for example in a GIS one. The scripts provide ways to visualize the different searches onto a map of the Netherlands. This was done to be able to quickly search and visualize the search results without having to export and then import files into a GIS environment.

Package name	Version number
certifi	2026.2.25
contourpy	1.3.3
cycler	0.12.1
fonttools	4.63.0
geopandas	1.1.3
kiwisolver	1.5.0
matplotlib	3.10.8
numpy	2.4.4
packaging	26.0
pandas	3.0.2
pandas-stubs	3.0.0.260204
pillow	12.2.0
pip	25.1.1
pyogrio	0.12.1
pyparsing	3.3.2
pyproj	3.7.2
python-dateutil	2.9.0.post0
scipy	1.17.1
shapely	2.1.2
six	1.17.0
tzdata	2026.1

Table 1: Overview of all the installed packages in the Python environment

5.2 Process description

5.2.1 Starting conditions

Provided with export of the current dataset (received April 13, 2026), which contained 5 csv files. The files were ‘01 – Sites’, ‘02 – Site_types’, ‘04 – Finds’, ‘03 – Codes’, and ‘06 – Codes_Sites’. These files had to be combined in a way to make one bigger dataset to work with within the python environment. Within this environment the files ‘01 – Sites’, ‘02 – Site_types’, and ‘04 – Finds’ were used.

5.2.2 Programming phase 1

The programming process was started with writing a prototype for the data processing and data visualization. This was done in native Python plus the utmost necessary packages. This prototype was as much a prototype of the code as it was a prototype of the analytical process. At this stage of the project the code more in support of exploring what could be done with the data, than actual code for the final product. This lead to the realisation that more advanced packages were needed to do spatial analysis, thus leading to the inclusion of the GeoPandas and Pandas packages.

5.2.3 QGIS phase 1

In QGIS a new project was started with coordinate system EPSG: 28992 Amersfoort / RD New. OpenStreetMap was added as the base layer (“OpenStreetMap”, n.d.). The 04 – Finds and 02 – Site_types csv’s were added via ‘add delimited text layer’. For the sites symbology was changed to different shape/size icons based on site type. For finds the symbology was based on material of the find, with different coloured icons for the different values. Having loaded the most basic of data in, it became clear that some of the sites were located outside of the border of the Netherlands. To be able to filter those out, a WFS layer was added with a connection to the ‘Gebiedsindelingen’ provided by the Centraal Bureau voor de Statistiek Centraal Bureau voor de Statistiek (CBS) (n.d.). A copy of this was exported as a GeoPackage to be able to import it into the Python environment. It was opted to proceed this way, as this did not have the information added from the 01 – Sites csv.

5.2.4 Programming phase 2

With the inclusion of the packages of Pandas, GeoPandas, and the required packages for those (see table 1) the next phase started. The csv files were now loaded into a Pandas dataframe. Within the Sites dataframe a new column was created in which Place (Cuijk) and Toponym (Maas) were combined into one site name (Cuijk-Maas). Sites and Site-types were then combined on SiteID, with the Place and Toponym columns removed.

The Finds data contained quite a few entries that did not have a specific sub-site attributed to them, as noted by the NULL in the Site_subID column. After inquiring with Groenhuijzen (personal communication) it was decided to treat these finds, and finds that had a zero in this column, as attributed to the entire site. This meant that a new sub-site code had to be created, and the code '-1' was chosen as to not interfere with the current data. A function was written, `site_null`, which retrieved all the unique Site_IDs and replace the Site_subID with the new number.

In the function `sites_cid` the Site_ID and Site_subID were combined into a combined site ID (Site_CID), with the separator of a decimal point (for example 1.-1). The coordinate columns were renamed to just X and Y (from X-coordinate_RD and Y-coordinate_RD) and the columns were reordered in such a way that SiteID and Site_CID would be the first two columns. The original Site data and the newly acquired entire site data were fed into this function and the results were combined into one dataframe containing all the different (sub)sites.

The sites dataframe was then converted into a geodataframe. A GeoPackage with the provincial boundaries of the Netherlands was also loaded in as a geodataframe. The geometries of the provinces were unionized and sites that were lying inside the country were selected. The sites dataframe with the remaining sites was then exported to a GeoPackage file for later use.

For the Finds data, a separate combined ID function was written, `finds_cid`. This was done for two reasons. First as the missing values and the value of zero needed to be changed into -1, and second because the Site_subID was stored as a float instead of an integer, thus needing an extra type conversion to be able to use the same format as the Site_CID outcome. This function also reordered the columns as to make Site_ID and Site_CID the first two columns. After running the Finds data through this combined ID function, only the finds that had a Site_CID that matched against a combined IDs in the Sites data were kept.

Search functions were written for both sites and finds. For sites one can search on three items. First on site name, `site_name`, as created above, second on site ID, `site_id`, and last on site type, `site_type`. For finds search queries were written for the find material, `find_material`, and find type, `find_type` columns. These functions return the sites where finds containing the input material/type are found. The sites returned from these are plotted on a map of the Netherlands for a visual representation. Each type of search has a different colour attributed to them, as to make plotting of different types possible.

5.2.5 Programming phase 3

Since all code to this point had been written in one continuous file, it was clear that for usability by others a restructure was needed. A separate *config.py* was created, in which the paths to the csv files and the provincial boundaries could be changed. This would mean that in later use, with different data, only the file paths in this one file needed to be changed instead of in multiple places throughout the code. The creation of the combined ID, reordering of columns and filtering of sites and finds was moved to *combining_data.py* while the search functions were moved to *searching.py*. The plot functionality was moved to *plotting.py* and updated to be its own function, `plotting`, as to not make the user comment lines in or out depending on their use case.

5.2.6 QGIS phase 2

The GeoPackage of the sites created in Python was loaded in, with symbology set to different icons for different site types. To be able to do spatial analysis, a grid had to be created. It was opted to create this grid based on the layer boundaries of the Gebiedsindelingen layer from CBS. The grid consisted of hexagons, as Birch, Oom & Beecham (2007, referenced in Gillings et al., 2020, pp. 435) noted that hexagons are preferred over squares due to the brain having less issues with preceiving made up patterns. A size of 3 km between a vertex and the second nearest vertex was chosen based on the case study of The English Landscapes and Identities project (EngLaId) (Gillings et al., 2020, pp. 437), as this provided a sufficient resolution for visualizations. While the EngLaId team refined their resolution to a grid of 1x1km squares to allow for reasonably robust statistical analysis, it was opted to not do this for this project as to keep processing/calculating times within reason.

The resulting hexagon layer was north-south wise longer than the Netherlands, so it got trimmed using the ‘select by location’ function within QGIS in such a way that only hexagons that intersected with the provinces were left. Using the analysis tools within QGIS the amount of finds and sites were counted within each

hexagon. Using the ‘add geometry attributes’ the area of a hexagon was determined. Since this function uses the coordinate system measurements, this area was in square meters, while square kilometres were wanted. This meant that the area had to be divided by 1 million (10^6). The field calculator was used to achieve this.

Lastly, a two new columns were added to the attribute table of the hexagons, one for the density of sites per square kilometre, the other of total finds per square kilometre. The values in these columns were calculated using the ‘field calculator’ and divide the number of site/finds per hexagon by the area. The calculating of density happened both for the total number of sites and finds, and for the different site types and find materials.

5.2.7 Programming phase 4

The calculating of the densities in QGIS was time intensive, mostly because the counting of points of interest in a hexagon took a long time (up to 45 minutes of counting total finds values). This led to the question if it was possible to do this entire process in a Python environment with the use of GeoPandas, and if so, if this would speed up processing times. To test this, the hexagonal grid layer in QGIS was exported as a GeoPackage and then imported into the Python environment. A function was written, **calculate_hexagon_density**, that then calculated the density of points of interest in the hexagons. Hexagons that had a density value of zero, were converted to contain a NaN value instead, as this would indicate GeoPandas to not plot these. Another function was written, **plot_hexagon_density**, to be able to plot these densities against a map of the Netherlands.

Since it might be of interest to plot the density of certain find materials versus the site types, once again, another function was written, **versus_plot**. This takes two dataframes, one for the find material and one for the site type, calculates the density of those and plots them against each other in a scatter plot with the site density on the x-axis.

5.2.8 Programming phase 5

Since percolation is the main technique used so far in spatial analysis, it seemed interesting to try to achieve this within Python as well, as the main package mentioned is written in R (Maddison & Schmidt, 2020). In *clustering.py* two functions were written. The first one, **search_radius**, takes the site ID from a starting site, and a radius (in meters) and returns a dataframe that contains all

the sites within that radius. The second function, **clustering**, takes these same inputs but continues searching for new sites in the radii of each site added to the cluster. Both functions have an option to visualize the result. The input variable `visualize` then needs to be set to `True`.

5.3 Output

All of the output will be available through GitHub at <https://github.com/Squirrelmonkey93/MasterThesis>.

The structure will be explained in the READ ME that is hosted on GitHub as well, as not not have this report open at all times. This file will also have sources of the files used accredited, while literary sources that some decisions are based on will have their credentials in this report. The output consists of two main parts, Python and QGIS, as being shown in the process description above (section 5.2). The Python part consists of a main folder and two sub-folders. The main folder contains the scripts written. In the sub-folder 'provided_csv' are the csv files stored that are used as the defaults in the scripts. In the sub-folder 'gpkgs' the used GeoPackages are stored. The QGIS part consists of a main folder with only one sub-folder. In the main folder the project is stored, while the subfolder contains all the GeoPackages.

Chapter 6

Results

6.1 The search and plotting functions

The search functions `site_name`, `site_id`, `site_type`, `find_material`, and `find_type` all give back a dataframe of sites that match the specified search input. The dataframe consists of the columns 'SiteID', 'Site_CID', 'Name', 'Site_type', 'X', 'Y', and 'geometry'. This can also be saved in a newly specified variable to be used in further processes.

The function plotting uses all of the above mentioned search functions and visualizes the results on a map of the Netherlands. The colouration of the site in this plot depends on which search function was used to locate the site. If a site was found using its name, it becomes pink on the plot. Sites found by site ID become red, while sites found by site type become green. Sites found based on the finds on them are coloured in the blue spectrum. Cyan for find material results, and blue for find type results.

On a combined plot as seen in figure 1, a site can be both of a specific site type (in this case rural settlement) and have a find that matched with the material type requested. In that case the more specific result overlaps the more general result, and this site would then become cyan on the plot.



Figure 1: Result of the plotting function which in turn uses the different search functions. All rural settlements are coloured green. The site found based on name is coloured pink, while the site found on ID is coloured red. Cyan sites are the result of searching using the find material, while blue sites are find types.

6.2 Density functions

The function `calculate_hexagon_density` calculates the density of sites based on site ID counts in a provided dataframe. This dataframe can be created by one of the search functions above. If `export` is set to `True`, the result will be exported to a GeoPackage. It is used in the functions `plot_hexagon_density` and `versus_plot`. Figure 2 is the result of running `plot_hexagon_density` using the complete Finds dataframe. Hexagons without any find in them, are not shown on the map for clarity. Figure 3 is the result of this function by using the complete Sites dataframe.

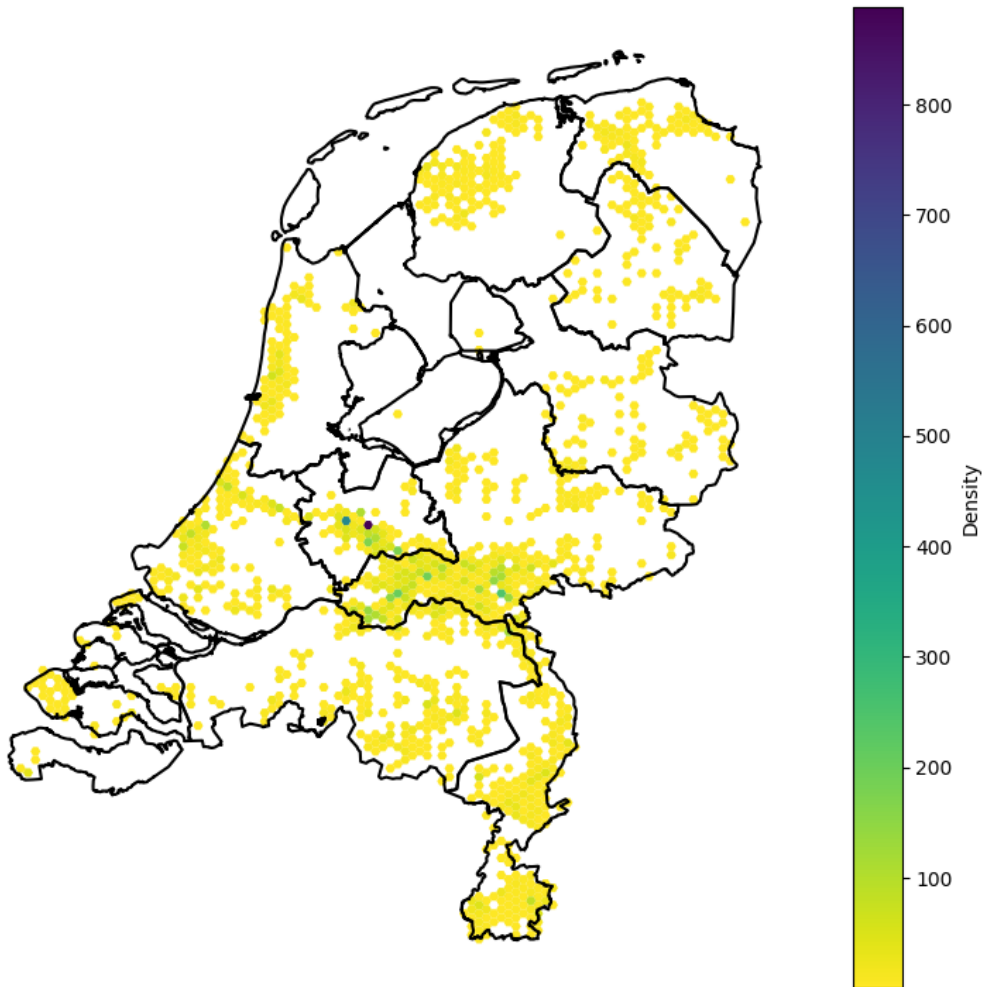


Figure 2: Density of the total Roman finds in the Netherlands. Hexagons without any find are not coloured in for clarity.

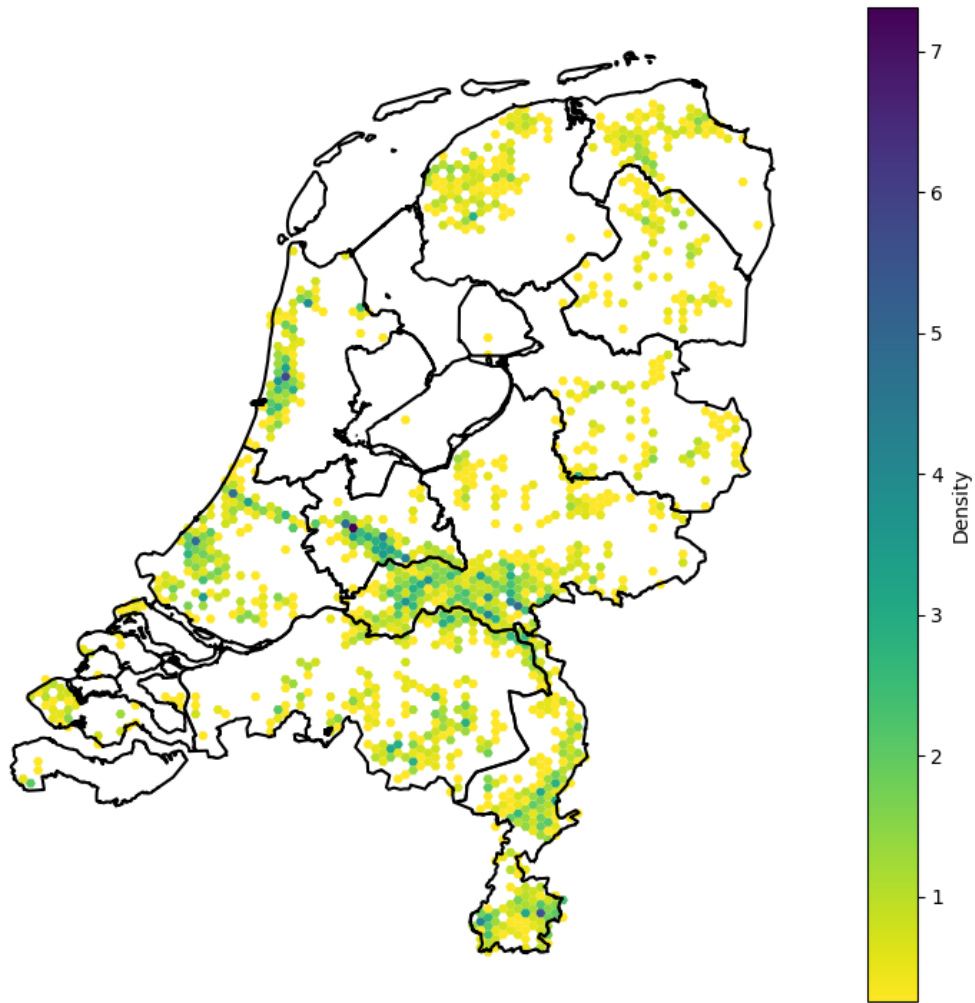


Figure 3: Density of all Roman sites in the Netherlands. Hexagons without any site in them are not coloured in for clarity.

The function **versus_plot** takes two dataframes, one on sites and one of finds and plots the densities of these against each other. Figure 4 show this plot for rural settlement sites and ceramic (keramiek) finds. The specific site type or specific material of a find can be changed by using the search functions above to create the dataframes for the input in this function.

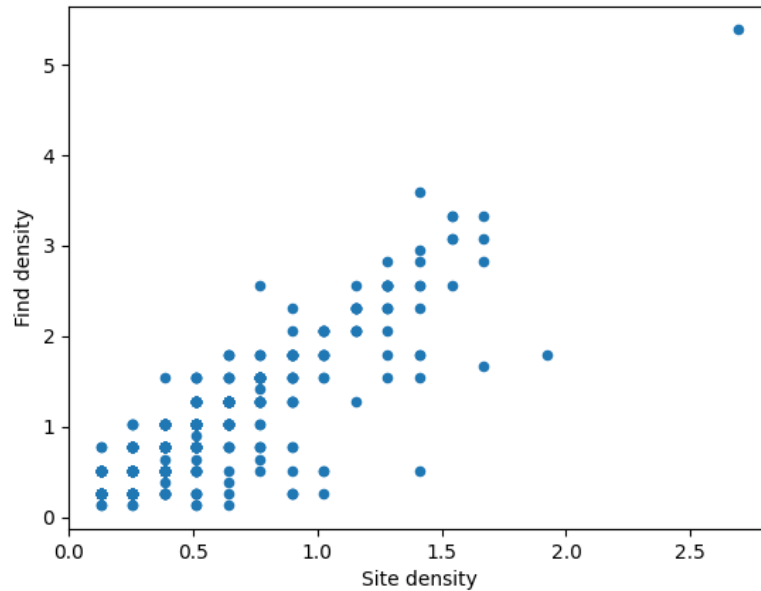
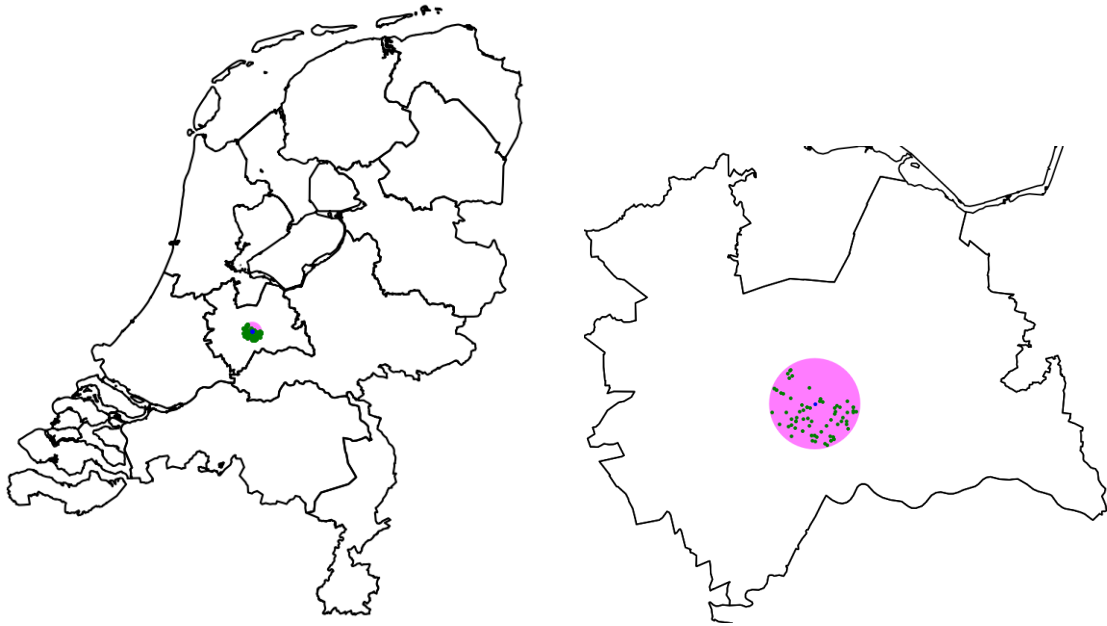


Figure 4: Site density of rural settlement type sites plotted against find density of ceramic (keramiek) finds.

6.3 Clustering functions

The function `search_radius`, with SiteID 502 (location Bunnik-Fort Vechten) and a radius of 5000 meters gives and visualization set to True gives back figure 5a. The selected site is coloured blue, while the sites within the search radius are coloured green. If the figure is zoomed in to only show the province of Utrecht where the site is located, we get figure 5b. The colours of the selected site and sites within the search radius are the same, but the pink colour is the radius itself visualized.



(a) Result on top of the entire Netherlands.

(b) Result zoomed in to the Utrecht province.

Figure 5: Visualization of the `search_radius` function. Chosen site ID is 502, which stands for the site Bunnik-Fort Vechten. Search radius is 5000 meters. Sites within that radius are coloured green. Search radius itself is coloured pink.

The function **clustering** with the same parameters as the **search_radius** function above (start site Bunnik-Fort Vechten, 5000 radius, visualization True) gives the result as shown in figure 6. Bunnik-Fort Vechten is shown as blue, while all the other sites that fall within this cluster are shown in green. The border of Roman Netherlands is traceable within this visualization as it followed the river Rhine.

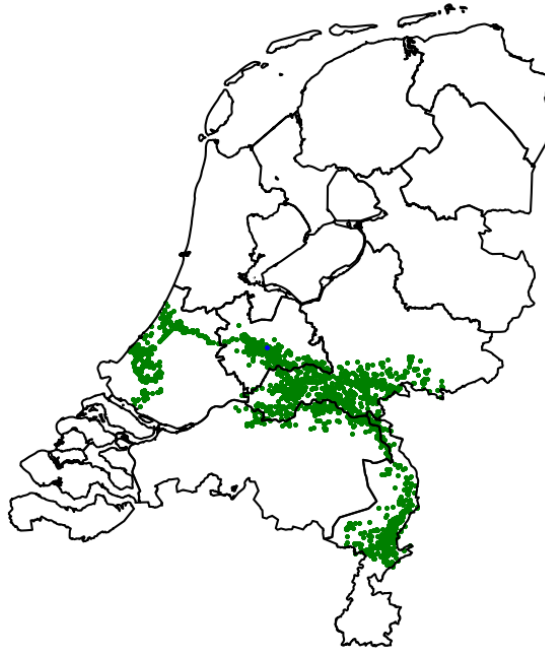


Figure 6: Visualization of the **clustering** function. Chosen site ID is 502, which stands for the site of Bunnik-Fort Vechten. Clusterings radius is 5000 meters. All the sites that fall within this cluster are colored green.

Chapter 7

Discussion

The aim of this project was to provide streamlined first step into spatial analysis of archaeological data while using data of the Netherlands in Roman times. Since site and find data are kept separately, combining these and creating visualizations based on specific criteria seemed like a good start into that. Being able to visualize what was found and where, on both a country wide scale, and on a smaller scale using the **search_radius** function, gives the potential for downsizing the data that need to be looked at for further research depending on the question asked.

A limited time frame to complete the entire project made it so that some elements are not done to their full potential. These elements are released as is for now, and could be build upon in the future. An example for this is the clustering function, as it only returns the cluster that the start site is in. Ideally there would also be a version of that, that starts a new cluster if no new sites could be found any more to add to the current one, and thus divide the entire dataframe provided into clusters. Another example is the filtering of the sites and finds. Right now, it is only possible to filter on site name, site ID, site type, find material, and find type. Ideally, a search function for time would also exist.

Besides these elements, there are also some that are missing entirely but would enhance the entire user experience. A dedicated space within the project do to statistical analysis is the one that seems to be the most obvious. It was started, but was let go due to the time constraints.

Another would be a better integration into a GIS environment. Creating, manipulating, importing, and exporting files between the GIS and Python environment should not have to take multiple windows and manually copy and pasting of items. This is partly the reason why the statistical analysis space was let go,

as the counting of selected sites within the palaeogeography gave back some errors.

The biggest item that would enhance user experience however, would be a proper user interface. Having drop down menus for selecting features to select by, or sliders to change time frame, would speed up the process even more as it does not require the future user get into the code and change things manually. This would also reduce the chance of user error because of spelling mistakes.

While the scripts are working for data exported from the Dutch ARCHIS database, it is most likely incompatible with exports of other countries. To improve the accessibility of this project the configuration file could either be updated or a new version for abroad data could be made, as to make the project also available for research outside of the Netherlands.

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